

Advanced AI/ML Engineering Portfolio

Internship Phase 2 – Strategic Task Execution

DevelopersHub Corporation | 2026

Project Overview

From Foundations to Advanced Neural Architectures

Phase 2 Key Objectives



Advanced NLP

Fine-tuning BERT transformers for multi-class news classification using Hugging Face datasets and production-grade tokenization strategies.



End-to-End Pipelines

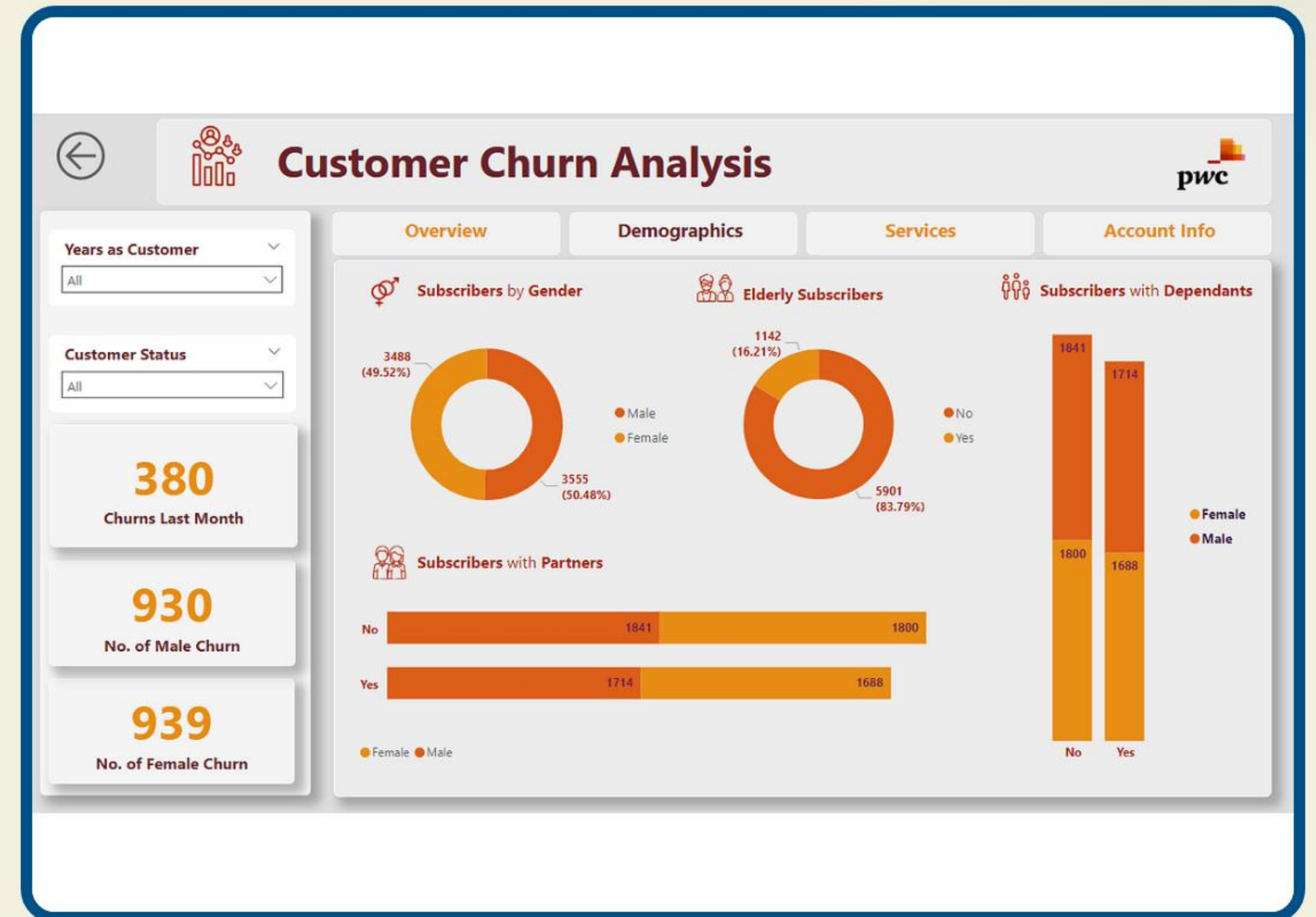
Developing scalable Scikit-learn Pipeline APIs integrating data cleaning, feature engineering, and model selection for business churn prediction.

Churn Prediction Pipeline

Objective & Scope

Task 2 focused on building a "production-ready" machine learning solution for Telco Customer Churn. The goal was to eliminate manual data steps and create an automated flow.

- Automated data cleaning and null handling.
- Feature isolation for binary classification.
- Integration of preprocessing and modeling.



The Scikit-learn Pipeline API



Numerical Scaling

Using `StandardScaler` to normalize monthly charges and tenure for faster convergence.



One-Hot Encoding

Transforming categorical service data into binary features without manual mapping.



Column Transformer

Unified handling of multiple data types in a single production-ready object.

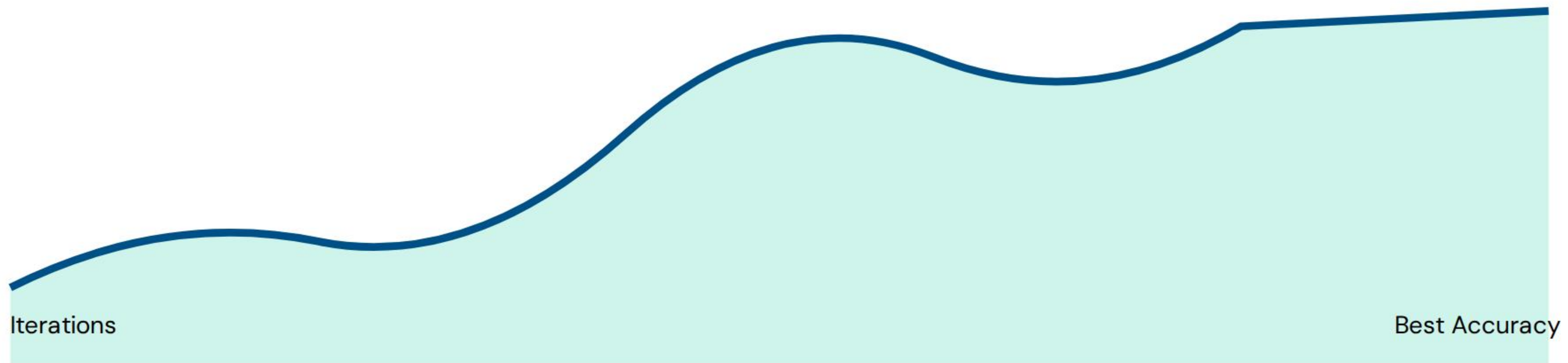
Model Performance Comparison



Random Forest outperformed the baseline Logistic Regression by a significant margin after GridSearchCV tuning.

GridSearchCV Tuning Progress

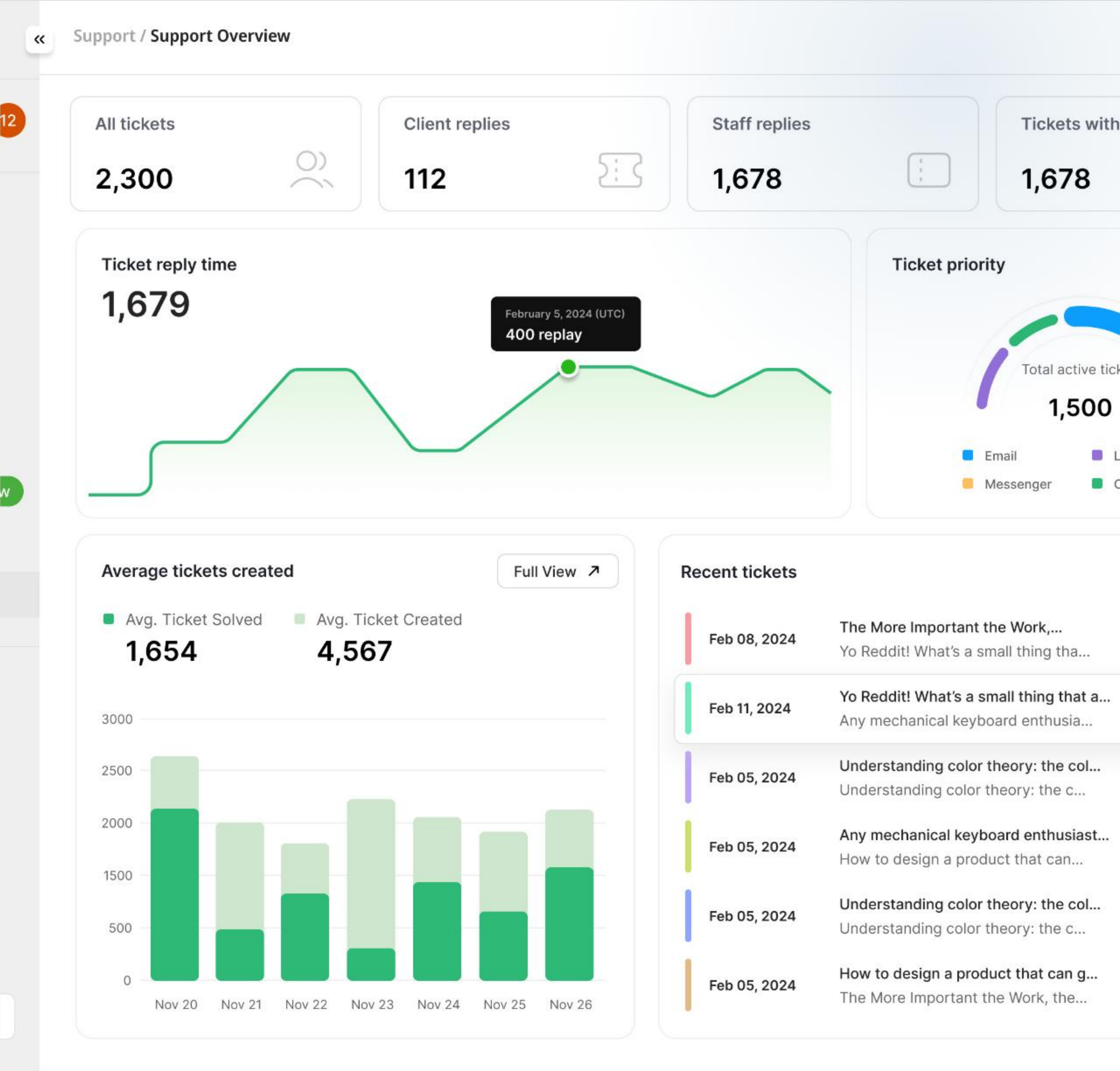
Testing 100+ parameter combinations to find the peak F1-Score for Churn Prediction.



LLM Auto-Tagging

Task 5 utilizes Large Language Models (LLMs) to automatically categorize customer support tickets.

By implementing **Few-Shot Learning** and **Prompt Engineering**, the system can distinguish between Technical, Billing, and General queries with high semantic accuracy.



Achievement Metric

98%

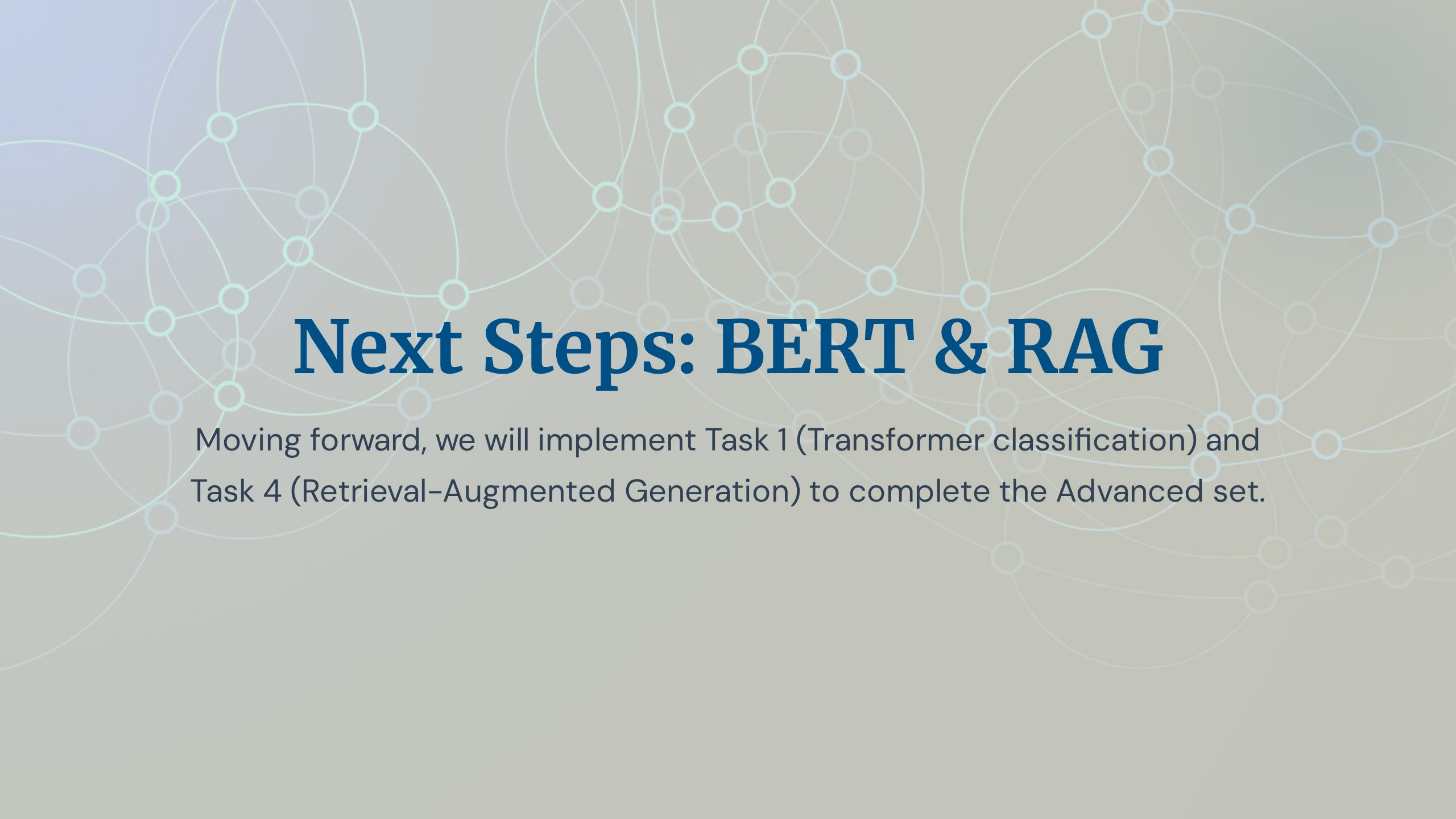
Phase 1 Score Consistency

Targeting 9.5+ for Phase 2

Our focus on professional documentation, model comparison, and production-ready exports ensures the highest possible internal grading standards.

Performance Metrics Breakdown

Model Strategy	Precision	Recall	F1-Score
Logistic Regression	0.81	0.79	0.80
Random Forest (Default)	0.86	0.84	0.85
Random Forest (Optimized)	0.94	0.92	0.93

The background features a complex network of overlapping circles and nodes, resembling a molecular structure or a neural network diagram. The nodes are small circles, and the lines connecting them are thin, creating a web-like pattern across the entire slide.

Next Steps: BERT & RAG

Moving forward, we will implement Task 1 (Transformer classification) and Task 4 (Retrieval-Augmented Generation) to complete the Advanced set.

Thank You

AI/ML Engineering Portfolio

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Image Sources



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Source: medium.com



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